Haroon Rasheed

AI & Automation Engineer | Python Developer | LLM & Workflow Automation

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PROFESSIONAL SUMMARY

AI and automation engineer who builds systems that businesses rely on every day. At Apex Innovation Technology I lead a small AI team and ship LLM features, n8n automations, and document intelligence pipelines that run in production. I work mostly with LangChain, the OpenAI API, FastAPI, PaddleOCR, and OpenVINO, focusing on robust, production-grade AI systems.

WORK EXPERIENCE

Junior AI Engineer — Lahore, Pakistan | Aug 2025 to Present
Apex Innovation Technology
- Lead a small AI team: plan roadmaps, conduct code reviews for juniors/interns before production release, and provide technical mentorship.
- Built five production n8n workflows for a US accounting firm categorizing transactions with an LLM, reconciling statements, and routing edge cases.
- Building HaeirLens / IDP Studio: FastAPI, PaddleOCR, and OpenVINO pipeline for appliance manufacturing; cut processing time to 15-20 seconds.
- Sole developer of enterprise AI Executive Assistant on Microsoft 365, and SAP Business One integrations with .NET and Angular.

Python Developer — Haripur, Pakistan | Jul 2024 to Oct 2024
Giga Developers
- Built Python backends with Flask and FastAPI, automating ML-driven data workflows and cutting manual processing time by 40%.

AI/ML Engineer Intern — Lahore, Pakistan | Aug 2024 to Sep 2024
EcodeCamp
- Engineered ML pipelines with TensorFlow and LangChain across three applied AI projects, boosting model accuracy by 25%.

Data Science Intern — India | Aug 2024 to Sep 2024
CodXo
- Developed and validated business forecasting models with Pandas, NumPy, and Scikit-Learn achieving 85% predictive accuracy.

Consumer Insights Business Analytics — California, USA | Aug 2024 to Sep 2024
Beats by Dre
- Conducted NLP sentiment analysis and competitive market intelligence to inform brand positioning strategy.

EDUCATION

Bachelor of Science, Artificial Intelligence
University of Haripur | 2021 to 2025 | Haripur, Pakistan

RESEARCH & CONTINUING EDUCATION

Internet Society Pre-Fellowship Training
Online Trust & Safety Track | Completed Oct 2026
Internet governance and online safety course graduate. Applying for the 2027 Internet Society Fellowship.

Conference Research Paper
Under Review | 2026 | ICAAD, Selcuk University
Submitted to the International Conference on AI Across Disciplines on applied vision and intelligence models.

TECHNICAL EXPERTISE

AI / LLMs: LangChain, OpenAI API, RAG, prompt engineering, agentic workflows, LangGraph, vector databases

Automation: n8n, workflow automation, webhook integration, QuickBooks Online integrations

Backend & APIs: Python, FastAPI, Flask, SQL, PostgreSQL, SQLAlchemy, JWT auth, Docker, .NET, Angular, SAP Business One

Computer Vision & OCR: PaddleOCR, OpenVINO, OpenCV, YOLOv8

Data & Deployment: Pandas, NumPy, Scikit Learn, TensorFlow, PyTorch, Git and GitHub

FEATURED PRODUCTION & OPEN SOURCE PROJECTS

AI Executive Assistant • *Internal product, in progress*
Multi-user Microsoft 365 copilot for email priority, task briefings, and Teams minutes via local Whisper. On-premise FastAPI, PostgreSQL, pgvector.

AI Smart Glasses for Visually Impaired • *Research project* — GitHub ↗
Final year research project with M. Muneeb Khan. YOLOv8 obstacle detection, OCR, GPS with dual English & Urdu audio guidance.

HaeirLens Document Extraction / OCR • *Client project*
FastAPI, PaddleOCR & OpenVINO document intelligence pipeline for major appliance manufacturer; reduced processing time to 15-20s.

Bilingual Purchase Order Extractor • *Client project*
Parses Arabic and English POs via three fallback tiers: Gemini vision, Groq over RapidOCR text, and offline parser.

PulsePoint Attendance Intelligence • *Client project*
Geofence-based attendance API detecting location fraud via mock GPS, VPN, and Wi-Fi BSSID validation checks.

SAP Telegram Approvals • *Client project*
FastAPI and Telegram Bot API microservice routing purchase orders and invoices to managers with inline one-click approval.

LeadIntel.AI: B2B Lead Enrichment • *Client project* — GitHub ↗
Identifies ERP footprint from LinkedIn and company domains. FastAPI backend, React/TS UI, Gemini with Claude Haiku fallback.

n8n QBO Bookkeeping Automation • *Client project*
Five production workflows for US accounting firm covering transaction categorization, reconciliation, and automated invoice mapping.

Intent Aware AI Support Chatbot • *Open source* — GitHub ↗
Zero-token regex intent classifier routing queries to a GPT-4.1-mini agent via LangChain behind a single webhook endpoint.

AI Voice to Email Agent • *Open source* — GitHub ↗
Voice-driven email assistant using Gemini 2.0 Flash audio transcription, Google Drive context search, and automated Gmail delivery.

Self Learning FAQ Chatbot • *Open source* — GitHub ↗
Semantic search FAQ chatbot on pgvector that automatically composes and embeds verified new answers whenever no match exists.

NeuraSense Emotion Recognition • *Open source* — GitHub ↗
Real-time speech emotion recognition system built on a custom CNN, Bidirectional LSTM, and Multi-Head Attention architecture.

Hammar AI Virtual Assistant • *Open source* — GitHub ↗
Desktop assistant with dual speech/text interface integrating Wikipedia, DuckDuckGo, and WolframAlpha knowledge engines.